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# Coordinated Scheduling and Energy Storage Planning for Grid-AIDC Systems Considering Heterogeneous Tasks

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***Abstract*—The rapid proliferation of Artificial Intelligence (AI) workloads imposes sustained operational pressure on distribution networks, necessitating the transformation of AI-centric Data Centers (AIDCs) into flexible grid-interactive assets. However, existing planning frameworks rely predominantly on homogeneous load models that overlook the distinct resource requirements (e.g., GPU vs. CPU) and rigid Service Level Agreements (SLAs) of heterogeneous workloads. This modeling deficiency leads to resource mismatches and an overestimation of demand response potential. To address these challenges, we propose a coordinated framework for AIDC-Grid scheduling and Energy Storage System (ESS) planning. Specifically, a task-type-guided heterogeneous load model is first established to accurately quantify dispatchable flexibility and differentiate task shedding priorities. Then, a spatiotemporal dispatch strategy guided by two economic signals is developed to mitigate congestion risks during load shifting. Furthermore, a Clustering-Based Adaptive Multi-Objective Evolutionary Algorithm (CAMOEA) is introduced to solve the resulting non-convex optimization problem with irregular Pareto fronts. Case studies demonstrate that the proposed framework effectively coordinates spatiotemporal regulation with ESS buffering, correcting flexibility overestimation and ensuring critical service quality.**

***Index Terms*—AI-centric data center (AIDC), heterogeneous load model, energy storage planning, spatiotemporal dispatch, multi-objective optimization.**

## NOMENCLATURE

*Sets and Indices*

| | |
|---|---|
| $\mathcal{D}, d, d'$ | Set and index of AIDCs ($1 \dots N_{AIDC}$). |
| $\mathcal{B}, i, j$ | Set and index of all the buses ($1 \dots N_{bus}$). |
| $\mathcal{L}, ij$ | Set and index of lines connecting bus $i, j$. |
| $k$ | Index of specific sub-task types. |
| $\mathcal{T}, t$ | Set and index of time slots ($1 \dots T$). |
| $\mathcal{J}^{(\cdot)}$ | Set of type $(\cdot)$ tasks, where $(\cdot) \in \{tr,\ in,\ bat,\ itv\}$. |
| $\mathcal{J}^{flex}$ | Set of temporally flexible tasks. |
| $\mathcal{J}^{mig}$ | Set of spatially flexible tasks. |

*Parameters*

| | |
|---|---|
| $a_m^g, b_m^g$ | Affine coefficients of segment $m$ in the auxiliary-power model, $g \in \{AI, nonAI\}$ . |
| $B_d^{eff}, \overline{B}_d$ | Effective and nominal ingress bandwidth capacity of AIDC $d$. [Gbps] |
| $c_t^{comm}$ | Unit communication cost coefficient at time $t$. |
| $C_{d,k}^{in}, C_{d,k}^{itv}$ | Service capacities of task Type $k$. |
| $C_{s,k}^2$ | Squared coefficient of variation of task service time. |
| $\chi_k$ | Service-time variability factor, defined as $(1 + C_{s,k}^2)/2$. |
| $c_k$ | Task complexity of Type-$k$ inference queries. |
| $\overline{D}_k, \tau_d$ | Latency bound and transmission delay of task Type $k$.[s] |
| $D$ | Representative operating period used for annualization. [day] |
| $D_k^{in}, D_k^{itv}$ | Available queueing-delay bounds for Type-$k$ inference/interactive tasks. |
| $E_d^{min}, E_d^{max}$ | Minimum/maximum energy of ESS. [MWh] |
| $i_d$ | Represent AIDC $d$ is located at bus $i$ |
| $l_{ij}^{\max}$ | Thermal current limit of line $ij$. |
| $K^{ESS}$ | Unit cost coefficient for ESS capacity. |
| $\Lambda_{d,k,t}$ | Expected arrival rate of Type-$k$ inference tasks in interval $t$. |
| $M_{d,k}$ | Max physical resource quota for Type $k$ tasks. |
| $\hat{N}_{d,k,t}^{bat}$ | Number of batch tasks before temporal shifting. |
| $N_{d,k,t}^{TR}$ | Total number of AI training sub-tasks at $t$. |
| $M_d^{(\cdot)}$ | Number of servers, where $(\cdot) \in \{GPU, CPU\}$. |
| $\mathcal{M}^g$ | Set of affine segments for auxiliary-power model. |
| $P_{i,t}^{base}$ | Local base load at bus $i$. |
| $\hat{P}_{d,t}$ | Original power load of task $k$ at time $t$. [MW] |
| $P_i^{\max}$ | Maximum active power capacity of bus $i$.[MW] |
| $P_{srv}^{idle}, P_{srv}^{peak}$ | Idle/Peak power of servers (GPU/CPU). [W] |
| $P^{query}$ | Power consumption per inference task. [W] |
| $q_k^{in}, q_k^{itv}$ | Average payload size of migrated inference/interactive tasks.[Gb] |

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| | |
|---|---|
| $r$ | Discount rate of ESS investment. |
| $S^{avg}$ | Average electricity price. [$/MWh] |
| $S_t$ | Electricity price at time $t$. [$/MWh] |
| $s_k$ | Data scale of Type-$k$ inference queries. |
| $\sigma$ | Capital recovery factor of ESS investment. |
| $\Delta t$ | Scheduling interval length. [h] |
| $W^{in}_{d,k,t}, W^{itv}_{d,k,t}$ | Average response times. |
| $\alpha_k$ | Pricing coefficient converting the arbitrage index into economic value. |
| $\beta_k$ | Weighting coefficient converting the penalty index into monetary value. |
| $\eta$ | Charging/discharging efficiency of ESS. |
| $\hat{\lambda}^{in}_{d,k,t}, \hat{\lambda}^{itv}_{d,k,t}$ | Equivalent arrival rates of inference/interactive tasks before spatial scheduling in interval $t$. |
| $\bar{\lambda}^{in}_{d,k,t}, \bar{\lambda}^{itv}_{d,k,t}$ | SLA-admissible arrival/processing-rate thresholds. |
| $\hat{\lambda}_{k,t}$ | Aggregate equivalent arrival rate of Type $k$ tasks before spatial shifting. |
| $\mu_k$ | Service rate for task Type $k$. |
| $\Omega_k$ | Scheduling deadline window for task Type $k$.[h] |
| $\omega^{bus}, \omega^{line}$ | Weights for bus/line overload objectives. |
| $\omega^{in}_k, \omega^{itv}_k$ | Task shedding priority weights. |
| $Y$ | ESS lifespan. [year] |
| $\rho^B_d$ | Bandwidth derating factor of AIDC $d$. |

*Variables*

| | |
|---|---|
| $C^{ele}$ | Total electricity purchasing cost. |
| $C^{inv}$ | Total investment cost of ESS. |
| $C^{op}$ | Total operating cost |
| $C^{comm}$ | Total communication cost. |
| $C^{SMC}$ | Total spatial migration cost. |
| $E_{d,t}$ | Energy stored in ESS of AIDC $d$ at time $t$. |
| $l_{ij,t}$ | Squared current magnitude on line $ij$. |
| $m^{bat}_{d,k,t}, m^{itv}_{d,k,t}$ | Number of active servers processing batch / interactive tasks at time $t$. |
| $N^{(\cdot)}_{d,k,t}$ | Number of Type $(\cdot)$ tasks, where $(\cdot) \in \{tr, bat\}$. |
| $N_{d,k,t\to t'}$ | Number of temporally flexible tasks shifted from time $t$ to $t'$. |
| $P^{(\cdot)}_{d,k,t}$ | Power consumption of task Type $(\cdot)$, where $(\cdot) \in \{tr, in, bat, itv\}$. [MW] |
| $P^{AI}_{d,t}, P^{nonAI}_{d,t}$ | Power of AI/non-AI tasks in AIDC $d$ at $t$.[MW] |
| $P^{ch}_{d,t}, P^{dis}_{d,t}$ | Charging/Discharging power of ESS.[MW] |
| $P_{i,t}, Q_{i,t}$ | Total active/reactive power at bus $i$. |
| $P_{ij,t}, Q_{ij,t}$ | Active/reactive power flow on branch $ij$. |
| $P^{IT,g}_{d,t}$ | IT-side power consumption of workload group $g$ in AIDC $d$. [MW] |
| $P^{aux,g}_{d,t}$ | Auxiliary facility power of workload group $g$ in AIDC $d$. [MW] |
| $L^{(\cdot)}_{d,k,t}$ | Shed rate for Type $(\cdot) \in \{in, itv\}$. |
| $\varepsilon_{ij,t}, \varepsilon^{max}$ | SOCP relaxation gap and its maximum value. |
| $R^{TIS}$ | Total time incentive savings. |
| $R^{bus}, R^{line}$ | Overload index for bus/line. |
| $S^{ESS}_d$ | Optimized ESS capacity in AIDC $d$. |
| $SMC_{d,k,t}$ | Spatial migration cost generated by Type-$k$ task migration at time $t$. |
| $TIS_{d,k,t}$ | Time incentive savings generated by Type-$k$ temporal scheduling at time $t$. |
| $\theta_{d,k,t}$ | Congestion penalty factor associated with AIDC $d$ at time $t$. |
| $u^{ch}_{d,t}, u^{dis}_{d,t}$ | Binary status of ESS charging/discharging. |
| $v_{i,t}$ | Squared voltage magnitude at bus $i$ and time $t$. |
| $\zeta^{in}_{d,k,t}, \zeta^{itv}_{d,k,t}$ | Admitted processing rate of Type-$k$ inference/interactive tasks. |
| $\lambda_{d'\to d,k,t}$ | Number of tasks migrated from AIDC $d'$ to $d$. |
| $\lambda^{in}_{d,k,t}, \lambda^{itv}_{d,k,t}$ | Equivalent arrival rates of inference/interactive tasks after spatial scheduling in interval $t$. |

*Functions*

| | |
|---|---|
| $F(\cdot)$ | Sub-objective functions. |
| $HV$ | Hypervolume indicator. |

*Abbreviations*

| | |
|---|---|
| **CAMOEA** | Clustering-Based Adaptive MOEA. |
| **TF, SF** | Temporal and spatial flexibility. |
| **MOEA/D** | MOEA based on Decomposition. |
| **SMC** | Spatial Migration Cost. |
| **SOCP** | Second-Order Cone Programming. |
| **TIS** | Time Incentive Savings. |
| **TOU** | Time-of-Use electricity price. |

## I. Introduction

THE rapid proliferation of Large Language Models (LLMs) has driven the expansion of data center (DC) infrastructure, leading to a surge in global power consumption [1]. The International Energy Agency (IEA) forecasts that global DC electricity consumption could exceed 1000 TWh by 2030, with AI workloads accounting for approximately 20% of the total consumption [2]. Unlike traditional DCs, AI-centric data centers (AIDCs) are characterized by high-density power profiles and task-type-dependent flexibility profiles. These characteristics impose sustained pressure on the local grid infrastructure [3], exacerbating the risks of overloading buses and transmission lines. In response, hyperscalers like Google have announced plans to shift or reduce power for some AI workloads to alleviate local grid strain [4]. Consequently, transforming AIDCs from passive loads into grid-interactive assets [5] by accurately characterizing and leveraging their dispatch potential is essential for local grid security [6].

Establishing an accurate heterogeneous load model is a prerequisite for effectively utilizing AIDC flexibility. Conventional studies primarily characterize DC flexibility at an aggregate level, relying on binary classifications of delay-sensitive and delay-tolerant tasks within traditional non-AI environments [7]–[10]. Recent studies have further considered task-dependency-aware flexibility modeling [11], cross-regional dispatchable capacity evaluation [12], and SLA-constrained flexibility quantification [13]. Driven by the rise of generative AI, specialized efforts have also distinguished the

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power profiles of training and inference workloads [14], [15]. However, these studies usually focus on either conventional non-AI workloads or AI-specific clusters, failing to reflect the hybrid operational reality of modern AIDCs where AI and non-AI tasks coexist. These workloads exhibit fundamental heterogeneity in hardware requirements (GPU vs. CPU) and Service Level Agreements (SLAs). For instance, AI training requires sustained high-intensity GPU occupancy and offers significant temporal flexibility. In contrast, AI inference and interactive services demand strict latency guarantees on GPUs and CPUs, respectively [16]. Applying homogeneous models to such heterogeneous loads may lead to resource mismatches (e.g., misallocating CPU tasks to high-power GPUs), thereby causing operational inefficiency and a systematic overestimation of demand response potential. More importantly, existing frameworks cannot explicitly separate the rigid portion of the mixed workload from the truly schedulable region once task-specific resource occupancy and multi-level SLA constraints are enforced. Without this rigid-dispatchable boundary, the scheduling model may overestimate the flexibility available for grid interaction and lead to excessive shedding of critical workloads under resource scarcity. Therefore, a refined heterogeneous load model is essential to accurately identify dispatchable boundaries while ensuring service quality.

Beyond load heterogeneity, a second critical issue is how to leverage the spatiotemporal flexibility of AIDCs in a grid-aware and coordinated manner. Existing literature primarily characterizes Temporal Flexibility (TF) as temporal task shifting within deadline windows [8], [17], and Spatial Flexibility (SF) as geographical task shifting constrained mainly by IT bandwidth [9], [10]. However, these strategies are often limited to physical boundaries or IT-level constraints, while ignoring how the identified flexibility should be activated under local electrical stress and service constraints. As a result, uncoordinated load shifting or aggressive price arbitrage may redirect workloads toward already congested buses, thereby causing the over-exploitation of flexibility resources. Therefore, an ordered mechanism is needed to coordinate spatiotemporal flexibility while accounting for both economic incentives and electrical security constraints. Furthermore, relying solely on load scheduling is insufficient to mitigate extreme load fluctuations due to rigidity caused by strict task SLAs and physical resource constraints. This necessitates the integration of auxiliary resources such as Energy Storage Systems (ESS) as a supplementary buffer, despite their high deployment costs [18]. Although recent studies have integrated ESS into renewable energy frameworks and multi-energy portfolios [19]–[21], their focus remains predominantly on energy management or market participation. However, the complementary relationship between ESS and workload flexibility remains unclear. A lack of coordinated planning between flexibility resources and ESS capacity results in either over-reliance on costly storage or degraded service quality and grid security.

The resulting coordinated optimization model involves discrete decision variables (e.g., task shifting volumes) and heterogeneous SLA constraints, constituting a complex non-convex Multi-Objective Optimization Problem (MOP). The Pareto front of such problems is often irregular, fragmented, or degenerate. Traditional algorithms such as NSGA-II, which rely on crowding distance for diversity preservation [22], may struggle to maintain convergence and diversity on such discontinuous fronts [23]–[25]. Consequently, a decomposition or clustering-based approach capable of adapting to the irregular Pareto front is required for improving solution diversity and maintaining search effectiveness in complex decision space.

As summarized in Table I, existing studies have not fully connected heterogeneous AIDC task modeling, grid-aware coordination of spatiotemporal flexibility, and ESS planning under grid-security and service-quality constraints. Accordingly, the main contributions of this paper are as follows:

1) To address the lack of an explicit boundary between rigid and schedulable demand in aggregate AIDC models, a refined heterogeneous load model for AIDCs is established based on task-energy mapping to quantify dispatchable flexibility. As shown in Fig. 1, the model distinguishes the resource requirements and multi-level SLAs of AI and non-AI workloads, including training, inference, interactive, and batch tasks. Furthermore, it identifies the rigid and dispatchable regions hidden inside the aggregate AIDC load, thereby reducing the flexibility overestimation caused by homogeneous descriptions.
2) To support the orderly release of AIDC spatiotemporal flexibility under both economic and electrical limits, a dual-signal dispatch strategy is proposed. The Temporal Incentive Savings (TIS) index quantifies arbitrage benefits from Time-of-Use (TOU) prices and guides temporally flexible tasks toward off-peak periods. Simultaneously, the Spatial Migration Cost (SMC) index acts as a congestion pricing signal by penalizing potential bus overloads, thereby reducing the risk that uncoordinated spatial shifting increases overload at receiving buses.
3) A coordinated Grid-AIDC scheduling and ESS planning framework is established to characterize the complementary roles of workload flexibility and ESS buffering, thereby reducing reliance on storage-only regulation. To address the discontinuous Pareto front arising from discrete task shifting and heterogeneous constraints, a Clustering-Based Adaptive MOEA (CAMOEA) is introduced to improve solution diversity and provide representative planning alternatives on the irregular Pareto front.

The rest of the paper is organized as follows: Section II details the heterogeneous load model and economic indicators. Section III formulates the planning problem, solved by CAMOEA in Section IV. Finally, Sections V and VI present case studies and conclusions.

## II. Heterogeneous Task Model and Scheduling Strategy

### A. Heterogeneous task model via refined AIDC task-energy mapping

To capture the operational heterogeneity of AIDCs, a discrete-time model at the system level is adopted, denoted as $t \in \mathcal{T} = \{1, 2, \cdots, T\}$. Different workload dynamics are mapped into a unified discrete-time coordination model through task-specific boundaries (e.g., execution windows and

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TABLE I
COMPARISON OF REPRESENTATIVE STUDIES AND RESEARCH GAPS

| Research stream | Representative studies | Main focus | Research gap |
|---|---|---|---|
| Aggregate DC flexibility | [7]–[10], [12] | Cluster-level flexibility and workload shifting | Rigid and dispatchable regions are not explicitly separated under task-specific SLA and hardware constraints. |
| Task-aware AI workload modeling | [11], [13]–[15] | Task dependency, SLA constraints, and AI training/inference profiles | Mixed AI/non-AI workloads and GPU/CPU-specific flexibility boundaries are not fully characterized. |
| Spatiotemporal grid interaction | [8]–[10], [17] | Temporal shifting, spatial migration, and grid operation | Spatial and temporal flexibility is not coordinated under both economic incentives and receiving-bus security limits. |
| ESS / energy-resource planning | [19]–[21] | ESS, renewables, and multi-energy coordination | ESS sizing is not fully coordinated with heterogeneous workload rigidity and service-quality constraints. |
| Multi-objective solution methods | [22]–[25] | Evolutionary optimization and irregular Pareto fronts | Solution diversity may deteriorate when irregular fronts are induced by coupled task scheduling and grid constraints. |
| This work | – | Heterogeneous AIDC modeling, grid-aware flexibility coordination, and ESS planning | Addresses the above gaps within a unified Grid-AIDC scheduling and ESS planning framework. |

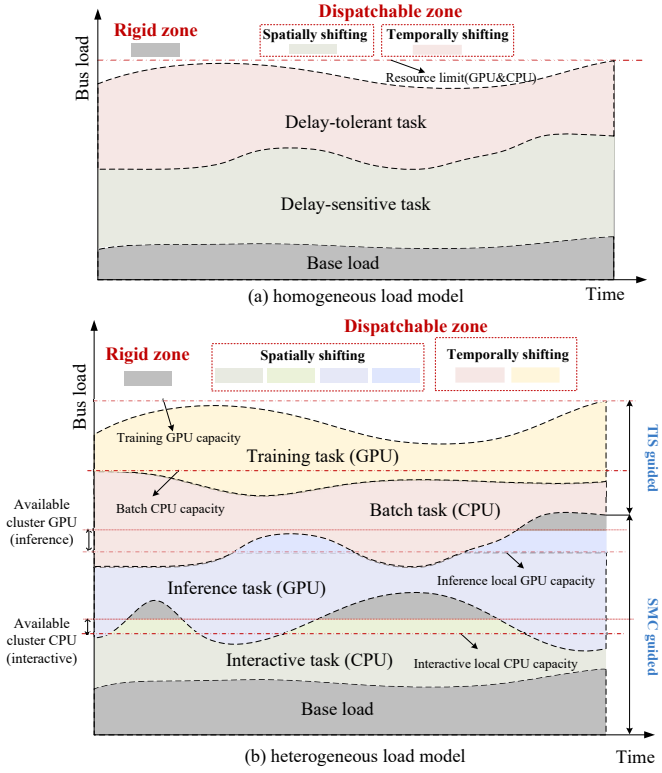



Fig. 1. Comparison of heterogeneous and homogeneous task models.

SLA constraints) for integrated grid and data center dispatch. For online tasks, the arrival and service processes are treated as piecewise stationary within each scheduling interval, so that queueing-based latency constraints can be embedded into the interval-based scheduling model. We consider a set of AIDCs, $d \in \mathcal{D} = \{1, 2, \ldots, N_{AIDC}\}$, managed by a single operator within the same electricity market environment. Based on the task-energy mapping considering distinct resource requirements (GPU/CPU) and SLAs, AIDC workloads are divided into four categories: AI training, AI inference, non-AI batch, and non-AI interactive tasks. For each category, the dispatchable flexibility is quantified as the feasible scheduling region bounded by physical resource limits and specific SLA constraints, specifically the deadline windows for training/batch tasks and the latency limits for inference/interactive tasks. Let $\mathcal{J}^{tr}, \mathcal{J}^{in}, \mathcal{J}^{bat}$, and $\mathcal{J}^{itv}$ denote the index sets of the corresponding task subtypes.

*1) AI training tasks:* AI training tasks are characterized by high computational intensity and long-duration execution, requiring exclusive GPU cluster occupancy. Unlike traditional batch tasks, a training job executes for days and exhibits high TF. Furthermore, checkpoint mechanisms allow millisecond-level suspension and flexible resumption, offering these tasks significant scheduling potential. In this study, a large-scale AI training job is decomposed into a set of sub-tasks that can be scheduled within its deadline window $[t, t + \Omega_k^{tr}]$. The total workload of a training job is quantified in GPU-hours (e.g., 5,000 GPU-hours), representing the product of the required server units and the execution duration [26]. Each sub-task represents a resource unit. Given that GPUs operate at sustained high utilization during training, the power consumption of training tasks $P_{d,k,t}^{tr}$ is proportional to the number of active server units $N_{d,k,t}^{tr}$:

$$P_{d,k,t}^{tr} = N_{d,k,t}^{tr} \cdot P_{GPU}^{peak}, \quad \forall d \in \mathcal{D}, k \in \mathcal{J}^{tr}, t \in \mathcal{T} \tag{1}$$

The scheduling of training tasks is formulated as a temporal redistribution of sub-tasks. Let $N_{d,k,t\to t'}^{tr}$ denote the number of sub-tasks shifted from the original arrival time $t$ to a new time $t'$. To ensure workload conservation, Eq. (2) ensures that the sum of shifted sub-tasks equals the original demand $N_{d,k,t}^{TR}$. Eq. (3) calculates the actual workload processed at time $t'$ as the aggregation of all sub-tasks shifted to this slot from various original times $t$. Furthermore, task execution is bounded by the physical resource capacity $M_{d,k}^{tr}$, as formulated in (4).

$$\sum_{t'=t}^{t+\Omega_k^{tr}} N_{d,k,t\to t'}^{tr} = N_{d,k,t}^{TR} \tag{2}$$

$$N_{d,k,t'}^{tr} = \sum_{t\in\mathcal{T}} N_{d,k,t\to t'}^{tr} \tag{3}$$

$$0 \le N_{d,k,t'}^{tr} \le M_{d,k}^{tr} \tag{4}$$

It should be noted that SF is not considered in the AI training task model, because transferring training checkpoints across AIDCs may involve large state data and non-negligible synchronization overhead.

*2) AI inference tasks:* AI inference tasks serve real-time user requests and exhibit significant stochasticity and time-varying intensity. Different from AI training tasks, inference tasks are usually latency-sensitive and must satisfy strict SLA constraints. Meanwhile, they have inherent spatial flexibility, enabling low-latency migration across AIDCs via optical fiber

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networks. Considering the heterogeneous energy profiles of varying inference types (e.g., text vs. image generation), a task-level energy model is adopted. The energy consumption of an inference query depends on the data scale $s_k$ and task complexity $c_k$. Assuming the original arrival process of Type-$k$ inference tasks follows a non-homogeneous Poisson process [27], the power model is characterized as Eq. (6):

$$\hat{\lambda}^{in}_{d,k,t} \sim \text{Poisson}(\Lambda_{d,k,t}) \tag{5}$$

$$\hat{P}^{in}_{d,k,t} = \hat{\lambda}^{in}_{d,k,t} P^{\text{query}}(s_k, c_k), \quad \forall d \in \mathcal{D}, k \in \mathcal{J}^{in}, t \in \mathcal{T} \tag{6}$$

where $\Lambda_{d,k,t}$ denotes the expected time-varying arrival rate of Type-$k$ inference tasks, and $P^{\text{query}}(s_k, c_k)$ denotes the average power contribution of a single Type-$k$ query within one scheduling interval.

To mitigate local computational congestion, spatial load balancing is applied by employing the low-latency properties of the network. Let $\lambda^{in}_{d'\to d,k,t}$ denote the number of inference tasks migrated from AIDC $d'$ to AIDC $d$ within interval $t$. The post-migration equivalent arrival rate is shown in Eq. (7). Eq. (9) and (10) ensure the conservation of the total arrival rate.

$$\lambda^{in}_{d,k,t} = \hat{\lambda}^{in}_{d,k,t} + \sum_{d'=1,d'\neq d}^{N_{AIDC}} \lambda^{in}_{d'\to d,k,t} - \sum_{d''=1,d''\neq d}^{N_{AIDC}} \lambda^{in}_{d\to d'',k,t} \tag{7}$$

$$\lambda^{\text{in}}_{d,k,t}, \hat{\lambda}^{\text{in}}_{d,k,t}, \lambda^{\text{in}}_{d'\to d,k,t}, \lambda^{\text{in}}_{d\to d'',k,t} \geq 0 \tag{8}$$

$$\sum_{d\in\mathcal{D}} \sum_{d'=1,d'\neq d}^{N_{AIDC}} \lambda^{in}_{d'\to d,k,t} = \sum_{d\in\mathcal{D}} \sum_{d''=1,d''\neq d}^{N_{AIDC}} \lambda^{in}_{d\to d'',k,t} \tag{9}$$

$$\sum_{d\in\mathcal{D}} \hat{\lambda}^{in}_{d,k,t} = \sum_{d\in\mathcal{D}} \lambda^{in}_{d,k,t} \tag{10}$$

To capture the variability of AI inference service times caused by different response lengths and task complexities, the latency constraint is modeled using an M/G/1 queue model [28]. The service capacity of AIDC $d$ for Type-$k$ inference tasks is defined as $C^{\text{in}}_{d,k} = M^{\text{in}}_{d,k}\mu_k$. Under the M/G/1 framework, let $S^{\text{in}}_{d,k}$ denote the random service time of Type-$k$ inference tasks. Its squared coefficient of variation is defined as $C^2_{s,k} = \text{Var}(S^{\text{in}}_{d,k})/(\mathbb{E}[S^{\text{in}}_{d,k}])^2$. According to the Pollaczek-Khinchine formula, the average response time of Type-$k$ inference tasks is approximated as Eq. (11) with the stability condition (12). To satisfy the SLA requirement, as formulated in Eq. (13), the response time plus transmission delay $\tau^{\text{in}}_d$ should remain within the latency bound $\bar{D}^{\text{in}}_k$. By defining $D^{\text{in}}_k = \bar{D}^{\text{in}}_k - \tau^{\text{in}}_d$ and $\chi_k = (1 + C^2_{s,k})/2$ which explicitly accounts for the variability of service times, the maximum allowable arrival rate $\bar{\lambda}^{\text{in}}_{d,k,t}$ under the SLA delay bound can be derived as Eq. (14). Eq. (14) indicates that higher service-time variability reduces the admissible arrival rate under the same delay requirement.

When the migrated inference workload exceeds the SLA-admissible processing capacity, a shedding mechanism is introduced to protect the response-time requirement of processed tasks. Consequently, the volume of shed tasks $L^{\text{in}}_{d,k,t}$, the actual processed task quantity $\zeta^{\text{in}}_{d,k,t}$, and the effective power consumption are constrained in (15)-(17).

$$W^{\text{in}}_{d,k,t} = \frac{1}{C^{\text{in}}_{d,k}} + \frac{\lambda^{\text{in}}_{d,k,t}(1 + C^2_{s,k})}{2C^{\text{in}}_{d,k}(C^{\text{in}}_{d,k} - \lambda^{\text{in}}_{d,k,t})} \tag{11}$$

$$0 \leq \lambda^{\text{in}}_{d,k,t} \leq C^{\text{in}}_{d,k} \tag{12}$$

$$W^{\text{in}}_{d,k,t} + \tau^{\text{in}}_d \leq \bar{D}^{\text{in}}_k \tag{13}$$

$$\bar{\lambda}^{\text{in}}_{d,k,t} = \frac{D^{\text{in}}_k (C^{\text{in}}_{d,k})^2 - C^{\text{in}}_{d,k}}{D^{\text{in}}_k C^{\text{in}}_{d,k} + \chi_k - 1}, \quad D^{\text{in}}_k C^{\text{in}}_{d,k} > 1 \tag{14}$$

$$\zeta^{\text{in}}_{d,k,t} + L^{\text{in}}_{d,k,t} = \lambda^{\text{in}}_{d,k,t} \tag{15}$$

$$0 \leq \zeta^{\text{in}}_{d,k,t} \leq \bar{\lambda}^{\text{in}}_{d,k,t}, \quad L^{\text{in}}_{d,k,t} \geq 0 \tag{16}$$

$$P^{in}_{d,k,t} = \zeta^{in}_{d,k,t} P^{\text{query}}(s_k, c_k) \tag{17}$$

*3) Non-AI batch tasks:* Non-AI batch tasks are CPU-intensive and latency-tolerant workloads. Unlike AI training tasks, which require sustained high concurrency over extended periods, non-AI batch tasks consist of short-duration, independent execution units with scheduling windows of several hours. This enables the system to flexibly schedule tasks during off-peak windows based on CPU availability, without requiring prolonged exclusive use of hardware resources.

$$N^{bat}_{d,k,t}, \hat{N}^{bat}_{d,k,t}, N^{bat}_{d,k,t''\to t}, N^{bat}_{d,k,t\to t'} \geq 0, \quad \forall d \in \mathcal{D}, k \in \mathcal{J}^{bat}, 1 \leq t'' \leq t \leq t' \leq T \tag{18}$$

$$N^{bat}_{d,k,t} = \hat{N}^{bat}_{d,k,t} + \sum_{t''=t-\Omega^{bat}_k}^{t-1} N^{bat}_{d,k,t''\to t} - \sum_{t'=t+1}^{t+\Omega^{bat}_k} N^{bat}_{d,k,t\to t'} \tag{19}$$

$$\sum_{t\in T} \sum_{t''=t-\Omega^{bat}_k}^{t-1} N^{bat}_{d,k,t''\to t} = \sum_{t\in T} \sum_{t'=t+1}^{t+\Omega^{bat}_k} N^{bat}_{d,k,t\to t'} \tag{20}$$

Leveraging temporal flexibility, batch tasks are scheduled within the delay window $\Omega^{bat}_k$. The actual workload $N^{bat}_{d,k,t}$ is derived by aggregating the net deferred tasks (inflow minus outflow) as shown in (19), subject to non-negativity (18) and total conservation (20). Finally, power consumption is modeled in (21)-(22) under server capacity constraints $M^{bat}_{d,k}$.

$$P^{bat}_{d,k,t} = (P^{peak}_{CPU} - P^{idle}_{CPU}) \frac{N^{bat}_{d,k,t}}{\mu_k} + P^{idle}_{CPU} m^{bat}_{d,k,t} \tag{21}$$

$$0 \leq m^{bat}_{d,k,t} \leq M^{bat}_{d,k} \tag{22}$$

*4) Non-AI interactive tasks:* Non-AI interactive tasks (e.g., Web services) rely on CPUs to process structured data. Characterized by logic-intensive operations but low computational density, these tasks exhibit load fluctuations driven primarily by user traffic. Similar to AI inference tasks, they offer spatial flexibility. As their spatial flow conservation constraints and SLA-based task shedding mechanism follow the same mathematical logic as AI inference tasks (7)-(16), the details are not repeated. Critically, the service times of interactive tasks are assumed to be exponentially distributed. So the service process is modeled by an M/M/1 queue ($C^2_{s,k} = 1$).

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The average response time is approximated as (23), and the maximum allowable arrival count is formulated in (24).

$$W^{itv}_{d,k,t} = \frac{1}{C^{itv}_{d,k} - \lambda^{itv}_{d,k,t}} \tag{23}$$

$$\bar{\lambda}^{itv}_{d,k,t} = C^{itv}_{d,k} - \frac{1}{D^{itv}_{k}} \tag{24}$$

Their power consumption is formulated in (25)-(26), while (27) and (28) enforce the aggregate GPU and CPU capacity constraints for all task categories.

$$P^{itv}_{d,k,t} = (P^{peak}_{CPU} - P^{idle}_{CPU})\frac{\zeta^{itv}_{d,k,t}}{\mu_k} + P^{idle}_{CPU} m^{itv}_{d,k,t}, \quad \forall d \in \mathcal{D}, k \in \mathcal{J}^{itv}, t \in \mathcal{T} \tag{25}$$

$$0 \le m^{itv}_{d,k,t} \le M^{itv}_{d,k} \tag{26}$$

$$\sum_{k \in \mathcal{J}^{tr}} M^{tr}_{d,k} + \sum_{k \in \mathcal{J}^{in}} M^{in}_{d,k} \le M^{GPU}_{d} \tag{27}$$

$$\sum_{k \in \mathcal{J}^{bat}} M^{bat}_{d,k} + \sum_{k \in \mathcal{J}^{itv}} M^{itv}_{d,k} \le M^{CPU}_{d} \tag{28}$$

### B. Spatiotemporal Scheduling Guided by Economic Signals

To promote the coordinated participation of AIDC flexibility resources in grid interaction, we propose a spatiotemporal scheduling strategy driven by two economic indicators: Time Incentive Savings (TIS) and Spatial Migration Cost (SMC).

*1) TIS:* Temporal scheduling is driven by TOU electricity price differentials. For the Type-$k$ temporally flexible tasks, the TIS quantifies the arbitrage benefit derived from task shifting. It is calculated based on the normalized price differential and the shifted task volume, weighted by an incentive coefficient $\alpha_k$. The TIS generated at time $t$ and the total revenue $R^{TIS}$ are calculated as (29)-(30):

$$TIS_{d,k,t} = \alpha_k \sum_{t' \in \Omega_k} (S_t - S_{t'})/S^{avg} \cdot N_{d,k,t \to t'}, \quad \forall d \in \mathcal{D}, k \in \mathcal{J}^{flex} = \mathcal{J}^{tr} \bigcup \mathcal{J}^{bat}, t \le t' \tag{29}$$

$$R^{TIS} = \sum_{d \in \mathcal{D}} \sum_{k \in \mathcal{J}^{flex}} \sum_{t=1}^{T} TIS_{d,k,t} \tag{30}$$

where $S^{avg}$ represents the average daily electricity price, and $\alpha_k$ is a pricing coefficient converting the arbitrage index into economic value. This indicator incentivizes AIDCs to leverage temporally flexible tasks for grid peak shaving and valley filling while adhering to the AIDC's physical constraints and the grid's operational safety thresholds.

*2) SMC:* Spatial task migration across AIDCs is defined as a high-cost emergency measure. This operation not only risks degrading QoS due to data transmission latency but may also exacerbate local congestion and overload risks. Therefore, to prevent unnecessary cross-regional task transfers, we define SMC as a congestion pricing signal. $\beta_k$ is a weighting coefficient converting the penalty index into monetary value. The SMC generated at time $t$ for migrating the Type-$k$ task from source AIDC $d'$ to receiving AIDC $d$, and the total system risk transmission cost $C^{SMC}$, are formulated as (31)-(33):

$$SMC_{d,k,t} = \beta_k \sum_{d' \ne d} \theta_{d,k,t} \cdot \frac{\hat{\lambda}_{d,k,t}}{\hat{\lambda}_{k,t}} \cdot D_k^{-1} \cdot \lambda_{d' \to d,k,t} \quad \forall d \in \mathcal{D}, k \in \mathcal{J}^{mig} = \mathcal{J}^{in} \bigcup \mathcal{J}^{itv}, t \in \mathcal{T} \tag{31}$$

$$\theta_{d,k,t} = 1 + \hat{P}_{d,t}/P^{\max}_{d} \tag{32}$$

$$C^{SMC} = \sum_{d \in \mathcal{D}} \sum_{k \in \mathcal{J}^{mig}} \sum_{t=1}^{T} SMC_{d,k,t} \tag{33}$$

where $\theta_{d,k,t}$ represents the congestion penalty factor of the bus connected to AIDC $d$. As defined in (32), it is positively correlated with the ratio of the AIDC's baseline total bus load $\hat{P}_{d,t}$ to the bus capacity $P^{\max}_{d}$. A higher congestion level results in a larger SMC penalty, thereby preventing blind task migration toward heavily loaded buses. To mitigate service centralization risks, the ratio $\hat{\lambda}_{d,k,t}/\hat{\lambda}_{k,t}$ introduces a spatial concentration penalty, increasing inflow costs for AIDCs that already host a large portion of the Type-$k$ spatially flexible tasks. The term $D_k^{-1}$ indicates the inherent inflexibility.

Guided by the two economic signals (SMC/TIS), the system prioritizes utilizing TIS for temporal peak shaving and valley filling. Spatial scheduling is used when local resources cannot meet rigid SLA requirements.

## III. Coordinated Planning Framework

In this section, a coordinated ESS planning framework for Grid-AIDC considering task heterogeneity is proposed. Consider a radial distribution system represented by $\mathcal{G} = (\mathcal{B}, \mathcal{L})$. Let $i_d \in \mathcal{B}$ denote the index of the bus where the $d$-th AIDC is connected. Then a multi-objective optimization problem is formulated to determine the optimal ESS capacity and task dispatch strategies, aiming to minimize three sub-objectives: total cost, grid overload risk, and service violation risk.

### A. Objective Functions

*1) Total cost:* The first objective $F_1$ minimizes the total cost, comprising the annualized ESS investment $C^{inv}$ and the operating cost $C^{op}$. In (34)-(38), $C^{inv}$ is calculated via the Capital Recovery Factor $\sigma$ (determined by ESS discount rate $r$ and lifespan $Y$), while $C^{op}$ integrates electricity costs $C^{ele}$, the communication costs $C^{comm}$, the $C^{SMC}$ from spatial scheduling, and the $R^{TIS}$ from temporal scheduling over a representative period $D$.

$$\min F_1 = C^{inv} + C^{op} \tag{34}$$

$$C^{inv} = \sigma K^{\text{ESS}} \sum_{d \in \mathcal{D}} S^{\text{ESS}}_{d}, \quad \sigma = \frac{r(1+r)^Y}{(1+r)^Y - 1} \tag{35}$$

$$C^{op} = \frac{365}{D}(C^{ele} + C^{comm} + C^{SMC} - R^{TIS}) \tag{36}$$

$$C^{ele} = \sum_{t \in \mathcal{T}} S_t \left[ \sum_{i \in \mathcal{B}} P_{i,t} \right] \Delta t \tag{37}$$

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$$C^{\text{comm}} = \sum_{t\in\mathcal{T}}\sum_{d\in\mathcal{D}}\sum_{\substack{d'\in\mathcal{D}\\ d'\neq d}} c_t^{\text{comm}}\Bigg[\sum_{k\in\mathcal{J}^{\text{in}}} q_k^{\text{in}}\lambda_{d'\to d,k,t}^{\text{in}} + \sum_{k\in\mathcal{J}^{\text{itv}}} q_k^{\text{itv}}\lambda_{d'\to d,k,t}^{\text{itv}}\Bigg]. \tag{38}$$

In (39), $P_{i,t}$ denotes the total load power at bus $i$, this value is derived from the combination of the local base load, the integrated AIDC workload, and the charging/discharging power of the ESS. Let $\mathcal{D}_i = \{d \in \mathcal{D} | i_d = i\}$ represent the set of AIDCs connected to bus $i$. Eq. (39) indicates that the power consumption of AIDC $d$ is incorporated into the total load of bus $i$ if the AIDC is connected to that bus. Furthermore, $P_{d,t}^{AI}$ and $P_{d,t}^{nonAI}$ represent the power consumption of AI and non-AI tasks, respectively. These values are derived by aggregating the task-energy mapping results of all heterogeneous sub-tasks and their corresponding auxiliary facility power (cooling and related support loads), as formulated in (40)-(43).

$$P_{i,t} = P_{i,t}^{base} + \sum_{d\in\mathcal{D}_i}(P_{d,t}^{AI} + P_{d,t}^{nonAI} + P_{d,t}^{ch} - P_{d,t}^{dis}) \tag{39}$$

$$P_{d,t}^{IT,AI} = \sum_{k\in\mathcal{J}^{tr}} P_{d,k,t}^{tr} + \sum_{k\in\mathcal{J}^{in}} P_{d,k,t}^{in} \tag{40}$$

$$P_{d,t}^{IT,nonAI} = \sum_{k\in\mathcal{J}^{bat}} P_{d,k,t}^{bat} + \sum_{k\in\mathcal{J}^{itv}} P_{d,k,t}^{itv} \tag{41}$$

$$P_{d,t}^{AI} = P_{d,t}^{IT,AI} + P_{d,t}^{aux,AI} \tag{42}$$

$$P_{d,t}^{nonAI} = P_{d,t}^{IT,nonAI} + P_{d,t}^{aux,nonAI} \tag{43}$$

By introducing an endogenous utilization-coupled facility-power model, $P_{d,t}^{aux}$ is bounded by a piecewise-affine lower envelope, ensuring it accurately scales with the scheduled IT load. The constraint is formulated as eq. (44).

$$P_{d,t}^{aux,g} \geq a_m^g P_{d,t}^{IT,g} + b_m^g, \forall g \in \{AI, nonAI\}, m \in \mathcal{M}^g \tag{44}$$

where $\mathcal{M}^g$ denotes the set of affine segments for workload group $g$. The parameters $a_m^g$ and $b_m^g$ are determined through the empirical auxiliary-power trends, capturing the nonlinear relationship between IT load and auxiliary power.

*2) Grid overload risk:* To quantify the grid overload risk imposed by massive AIDC integration, a grid overload index $F_2$ is formulated as (45). This index comprises a weighted sum of the bus overload index and the line overload index, aiming to minimize the extent to which actual loads exceed their rated safety thresholds.

$$\min F_2 = \sum_{t\in\mathcal{T}}(\omega^{bus}\sum_{i\in\mathcal{B}} R_{i,t}^{bus} + \omega^{line}\sum_{ij\in\mathcal{L}} R_{ij,t}^{line}) \tag{45}$$

For the bus component, the bus overload index $R_{i,t}^{bus}$ quantifies the normalized ratio by which the active power at bus $i$ $P_{i,t}$ exceeds the maximum active power capacity $P_i^{\max}$ as (46). For lines, assuming the overload criterion is based on the thermal stability limit, let $l_{ij}^{\max}$ be the safety threshold for the square of the current on line $ij$. As in (47), the line overload index $R_{ij,t}^{line}$ employs a square root operation to quantify the overload severity relative to the rated current magnitude.

$$R_{i,t}^{bus} = \max(0, P_{i,t}/P_i^{\max} - 1), \forall i \in \mathcal{B}, t \in \mathcal{T} \tag{46}$$

$$R_{ij,t}^{line} = \sqrt{\max(0, l_{ij,t}/l_{ij}^{\max} - 1)}, \forall ij \in \mathcal{L}, t \in \mathcal{T} \tag{47}$$

*3) Service violation risk:* Under resource-constrained conditions, guaranteeing the QoS for critical tasks is important. As shown in (48), to quantify the risk of service violations, the index $F_3$ calculates the ratio of the shed volume $L_{d,k,t}$ to the original demand $\hat{\lambda}_{d,k,t}$. The weight parameters $\omega_k$ are introduced to define service priorities. Specifically, critical tasks with rigid SLAs are assigned higher weights, thereby resulting in high penalty costs to drive priority resource allocation. This mechanism ensures hierarchical protection under resource constraints, preventing the shedding of critical tasks. The shedding ratio is zero when task arrival ($\lambda_{d,k,t} = 0$).

$$\min F_3 = \sum_{d\in\mathcal{D}}\sum_{t\in T}\Bigg(\sum_{k\in\mathcal{J}^{in}} \omega_k^{in}\cdot L_{d,k,t}^{in}/\hat{\lambda}_{d,k,t}^{in} + \sum_{k\in\mathcal{J}^{itv}} \omega_k^{itv}\cdot L_{d,k,t}^{itv}/\hat{\lambda}_{d,k,t}^{itv}\Bigg) \tag{48}$$

## B. Constraints

The AIDC resource and task spatiotemporal transfer balance constraints are introduced in Section II, as detailed in Eq. (1)-(33). The remaining constraints are listed as follows.

*1) Bandwidth constraints:* To ensure the communication feasibility of spatial migration, the aggregate ingress traffic caused by migrated online workloads should not exceed the effective ingress bandwidth $B_d^{eff}$. As formulated in (49), $B_d^{eff}$ is obtained by derating the nominal ingress bandwidth capacity $\bar{B}_d$ to reserve part of the nominal capacity for background traffic, congestion fluctuation, and possible transfer failures, thereby providing a conservative migration boundary.

$$\sum_{d'\neq d}\left(\sum_{k\in\mathcal{J}^{\text{in}}} q_k^{\text{in}}\lambda_{d'\to d,k,t}^{\text{in}} + \sum_{k\in\mathcal{J}^{\text{itv}}} q_k^{\text{itv}}\lambda_{d'\to d,k,t}^{\text{itv}}\right) \leq B_d^{\text{eff}}\Delta t \tag{49}$$

$$B_d^{eff} = \rho_d^B \bar{B}_d, \quad 0 < \rho_d^B \leq 1 \tag{50}$$

*2) ESS operational constraints:* To prevent physically infeasible simultaneous charging and discharging, binary variables are constrained by $u_{d,t}^{ch} + u_{d,t}^{dis} \leq 1$. Moreover, the charging and discharging powers are limited by the ESS installation capacity, as formulated in (51).

$$0 \leq P_{d,t}^{ch} \leq u_{d,t}^{ch} P_d^{ch,\max}, 0 \leq P_{d,t}^{dis} \leq u_{d,t}^{dis} P_d^{dis,\max} \tag{51}$$

The energy state $E_{d,t}$ follows temporal continuity principles: $E_{d,t} = E_{d,t-1} + \eta P_{d,t}^{ch}\Delta t - P_{d,t}^{dis}\Delta t/\eta$, $E_d^{\min} \leq E_{d,t} \leq E_d^{\max}$, where $E_{d,0} = E_{d,T}$ ensures periodicity.

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*3) Power flow constraints:* The network constraints are modeled using the Second-Order Cone Programming (SOCP) relaxation of DistFlow branch equations [29], encompassing active/reactive power balance, voltage drops, and second-order cone constraints. Since standard SOCP relaxations are inherently prone to inexactness when power systems operate near boundary limits or when weighted-sum multi-objective functions are optimized [30], a small penalty term related to the relaxation gap is included in the operational scheduling problem to improve the numerical tightness of the SOCP relaxation. The tightness of the SOCP relaxation is verified a posteriori by evaluating the relaxation gap of the obtained solutions in Section V.

## IV. Solution Method

### A. Motivation for Adopting CAMOEA

The proposed planning problem has a nested structure: the outer layer determines the ESS capacity, while each candidate solution requires solving an inner operational scheduling problem involving the discrete decision variables (e.g., ESS binary states, task shedding), multi-level SLA constraints and SOCP-based network constraints. These coupled mechanisms make the mapping from ESS capacities to objective values highly non-smooth and create extensive infeasible regions. Consequently, the resulting Pareto front is not expected to be regular or uniformly distributed; instead, it typically exhibits discontinuity, fragmentation, and partial degeneration. Common Multi-Objective Evolutionary Algorithms (MOEAs), such as the Decomposition-based algorithm (MOEA/D), rely on a set of predefined, uniformly distributed weight vectors to guide the search [31]. However, fixed weight vectors may direct the population towards these invalid regions, leading to wasted computational resources and a loss of population diversity, as the algorithm struggles to approximate a PF that does not align with the preset geometric distribution.

Therefore, CAMOEA is adopted because its clustering-based environmental selection adapts to the actual distribution of nondominated solutions. In contrast, CAMOEA employs a clustering-based adaptation mechanism that dynamically generates and adjusts reference vectors based on the actual distribution of nondominated solutions. The cluster centroids then guide environmental selection, allowing the algorithm to maintain diversity even when the feasible front is discontinuous or non-uniform. This property matches the structural characteristics of our AIDC-grid scheduling problem.

### B. Computation steps

The proposed coordinated planning framework is solved via the following steps. The main problem is optimizing the ESS configuration and the subproblem handles the spatiotemporal task scheduling.

*1) Initialization:* Set the population size $N$, maximum generations $G^{\max}$, and number of objectives $M$. An initial population $P$ is randomly generated, where each individual represents a decision vector for ESS capacity ($\boldsymbol{S}^{\text{ESS}}$). The main evolutionary parameters are set as $N = 60$, $G^{\max} = 70$, crossover probability $p_c = 1$, mutation probability $p_m = 1/n$ (where $n$ is the number of decision variables), and the distribution indices for SBX and polynomial mutation are set to 20 [32]. Specifically, $N$ and $G^{\max}$ are selected to balance the diversity of Pareto sets and computational burden.

*2) Operational Simulation and Performance Assessment:* For each candidate solution in the population, an operational simulation is triggered to evaluate its performance.

- Subproblem Formulation: Given the candidate $\mathbf{S}^{\text{ESS}}$, the subproblem optimizes task scheduling and ESS operation.
- Solving and Feedback: CPLEX 12.10 is used to solve the subproblem. If a feasible dispatch scheme exists, the optimal scheduling scheme ($N^*_{d,k,t\to t'}$, $\lambda^*_{d'\to d,k,t}$, $P^{ch*}_{d,t}$, $P^{dis*}_{d,t}$, ...) is extracted, and the corresponding objective values are calculated (Total cost $F_1$, Grid overload risk $F_2$, and Service violation risk $F_3$). Feasible solutions yield objective values $(F_1, F_2, F_3)$, while infeasible ones (e.g., due to power flow divergence) are penalized.

*3) Evolutionary Optimization via CAMOEA:* Based on the fitness values obtained from the simulation, the planning variables are updated using the CAMOEA strategy, which involves the following steps:

- Evolutionary Operations: Offspring $Q$ is generated via Simulated Binary Crossover (SBX) and polynomial mutation to explore the investment decision space. Then combined with parents $P$ to form $R = P \cup Q$.
- Clustering-Based Selection: Non-dominated sorting is performed on $R$. Then, hierarchical clustering is applied to the irregular front. By adaptively identifying cluster centers, the best $N$ individuals for the next generation $P_{\text{new}}$ are selected based on diversity and convergence.

*4) Termination and Output:* The procedure repeats Steps 2 and 3 until the termination criterion (e.g., reaching $G^{\max}$) is met. Finally, the algorithm outputs the set of Pareto optimal ESS planning schemes.

As a heuristic approach, it cannot guarantee strict global optimality with a proven optimality gap compared to exact solvers. Its clustering procedure and repeated fitness evaluations also increase the computational cost, although the evaluations of individuals within each generation are independent and can be performed in parallel. The framework is intended mainly for offline ESS planning, where day-ahead operating profiles are incorporated to improve the evaluation of candidate planning schemes. It is not intended for real-time and millisecond-level control.

## V. Case Study

The proposed framework is evaluated on the modified IEEE 33-bus system. Three AIDCs, each equipped with an ESS, are integrated into the grid at bus 2, 7, and 19 (connection indices $i_1 = 2, i_2 = 7, i_3 = 19$), with maximum power capacities of 9.5 MW, 8.5 MW, and 11.2 MW, respectively. The simulation covers a representative period of 72 hours ($D = 3$ days) with results subsequently annualized for economic assessment. GPU servers operate with peak power of 450W per unit [33], CPU servers have idle/peak power of 100/200W per unit [34]. The numbers of GPU servers $M_d^{GPU}$ across the three AIDCs are 15,000, 13,200, and 15,000, respectively.

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Similarly, the CPU server counts $M_d^{CPU}$ are 7800, 7500, and 8100. Regarding the four categories of tasks within the AIDC, AI training tasks include three different types requiring computational resources of 4,000, 15,000, and 40,000 GPU-hours, respectively; AI inference tasks comprise two types, with power consumptions $P^{\text{query}}$ of 2.84 W and 3.96 W, respectively, to represent the average power contribution per query of standard ChatGPT and complex Bloom models [3]. The specific SLA constraints, including equivalent service rates and deadline windows, are summarized in Table II.

TABLE II
PARAMETERS AND SLA CONSTRAINTS FOR HETEROGENEOUS TASK CATEGORIES

| Task category | Task sub-type | Deadline window/SLA | Equivalent service rate ($\times 10^3$) |
|---|---|---|---|
| training | Type 1 | 24 h | 1.0 |
| | Type 2 | 72 h | 1.0 |
| | Type 3 | 48 h | 4.0 |
| inference | Type 1 | 0.15 s | 8.0 |
| | Type 2 | 0.10 s | 19.6 |
| batch | Type 1 | 6 h | 40.0 |
| | Type 2 | 4 h | 25.6 |
| interactive | Type 1 | 2.00 s | 60.0 |
| | Type 2 | 1.00 s | 40.0 |

For the AI-task data, we use a trace taken from the LLM services in Azure with real-world requests and timestamps [35]. Non-AI tasks are sourced from normalized workload traces of Google data centers [36]. Electricity pricing data is obtained from [37]. All case studies were implemented in MATLAB R2022b, and the embedded operational optimization problems were solved using CPLEX 12.10 on a computer equipped with an Intel Core i5-10200H CPU at 2.40 GHz and four physical cores. The following cases are tested to evaluate the effectiveness of the proposed model.

$Case\ A$: Comparative analysis of scenarios with and without the proposed economic indicators (TIS/SMC) to verify their role in guiding orderly spatiotemporal scheduling;

$Case\ B$: Comparison between the proposed heterogeneous task model and the traditional homogeneous model. This aims to demonstrate the necessity of task classification in preventing resource misallocation and avoiding critical SLA violations;

$Case\ C$: Evaluation of independent versus coordinated regulation strategies involving ESS and AIDC flexibility to verify how their coordination improves the trade-off among cost, grid security, and QoS;

$Case\ D$: Comparison of CAMOEA against MOEA/D to verify its advantages in convergence and diversity for solving complex non-convex multi-objective problems;

$Case\ E$: Evaluation of robustness and feasibility under practical operating conditions to verify the forecast-error robustness, PV-integration compatibility, and SOCP relaxation tightness of the proposed framework;

$Case\ F$: Sensitivity analysis of key economic parameters to investigate the effects of $\alpha_k$ and $\beta_k$ on the three objectives and identify recommended parameter ranges.

### A. Analysis of Economic Guidance Indicators.

Based on the proposed Grid-AIDC coordinated framework, a Pareto front with 60 optimal solutions is obtained as shown in Fig. 2. We apply the Fuzzy Multiple Attribute Decision Making (FMADM) [38] method to select a compromise solution from the Pareto front. The corresponding optimal ESS capacities for AIDC 1, 2, and 3 are configured as 12.00 MWh, 0.79 MWh, and 0.31 MWh, respectively. AIDC1 (Bus 2) requires a larger storage buffer because it is connected to the upstream bus with heavier baseline loading, whereas AIDC3 (Bus 19) can manage its power balance mainly through internal scheduling and available local grid margin.

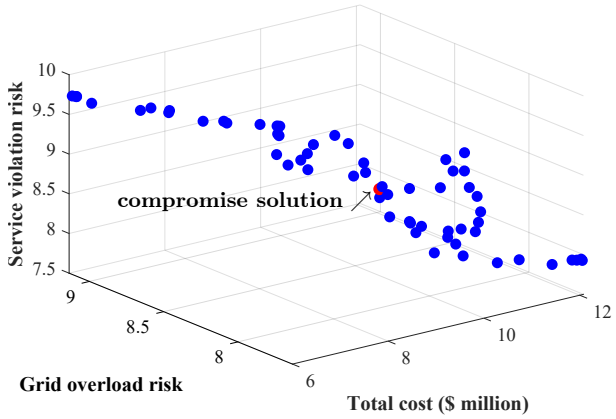



Fig. 2. Pareto solutions for the proposed MOP.

Fig. 3 shows how the TIS signal releases temporal flexibility. For the Type 3 training tasks in AIDC2, part of the workload is shifted from high-price periods to lower-price periods within the allowable deadline window. Although the completion ratio within the 72-h observation window decreases from 80.0% to 47.2% after temporal shifting, it does not imply service violation; rather, it reflects an economically guided deferral of flexible subtasks within their allowable deadlines. The TIS values in Table III further quantify this behavior. For example, AIDC2 obtains a TIS revenue of 24.2 kUSD for Type-3 training tasks, confirming that TIS converts deadline flexibility into active economic optimization.

TABLE III
ANNUAL AVERAGE ECONOMIC INDICATORS (TIS & SMC) FOR HETEROGENEOUS TASKS (KUSD)

| Location | TIS (AI training tasks) | | | SMC (AI inference tasks) | |
|---|---|---|---|---|---|
| | Type 1 | Type 2 | Type 3 | Type 1 | Type 2 |
| AIDC1 | 18.9 | 35.4 | 93.6 | 81.3 | 41.7 |
| AIDC2 | 28.1 | 45.7 | 24.2 | 205.5 | 193.8 |
| AIDC3 | 18.1 | 29.4 | 116.2 | 25.9 | 1.5 |

Fig. 4 illustrates the spatial migration of Type 1 inference tasks on Day 2. The gray curve shows that, before spatial shifting, task shedding occurs intermittently and exhibits several sharp peaks, indicating temporary local GPU service-capacity shortages under strict latency constraints. After spatial shifting, the black dashed curve remains close to zero for most time slots, demonstrating that inter-AIDC migration effectively absorbs these short-term service deficits. The bar components further reveal that spatial regulation is an emergency measure only performed during local congestion periods. At time slots such as $t = 6$ and $t = 17$, no transfer occurs as local resources are sufficient. However, substantial task transfers (e.g., $t = 2, 13, 18$) aim to compensate for the correspond-

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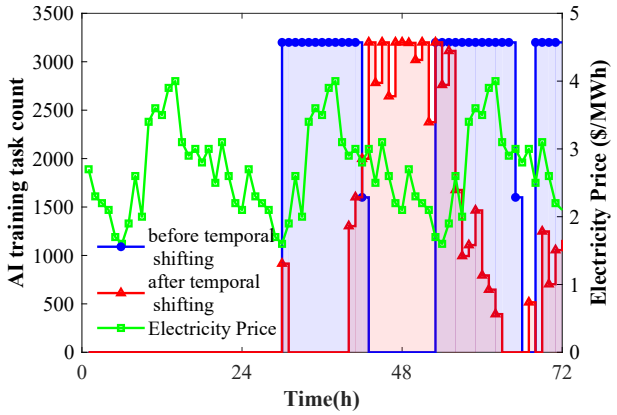



Fig. 3. Temporal scheduling profiles of Type 3 AI training tasks in AIDC2 before and after temporal shifting.

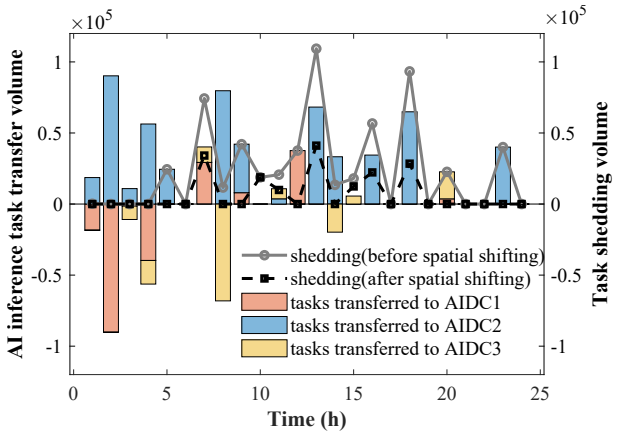



Fig. 4. Spatial allocation and shedding volume of Type 1 AI inference tasks among AIDCs on Day 2.

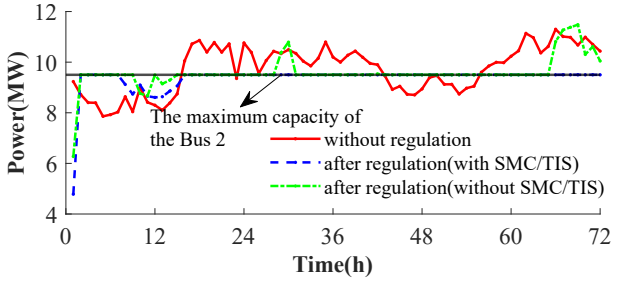



Fig. 5. Load power before/after regulation at Bus 2.

ing shedding peaks, thereby protecting services that would otherwise be lost. As shown in Table III, AIDC2 incurs the highest SMC due to large inflows, validating SMC's role as a congestion pricing signal that prevents excessive transfers that might stress the receiving bus. Finally, the necessity of SMC/TIS is validated by the comparative results in Figs. 5 and 6. Fig. 5 compares the load profiles at Bus 2 (AIDC1) under different regulation strategies. With economic signals, the bus load is effectively constrained below the capacity limit for most periods. Conversely, in the absence of SMC/TIS, the system relies solely on physical server capacity constraints without sufficient consideration of receiving-bus congestion. This leads to blind task migration: spatially flexible tasks surge into AIDC1 between t=66h and 72h, the overload magnitude in certain periods even exceeds the un-regulated benchmark.

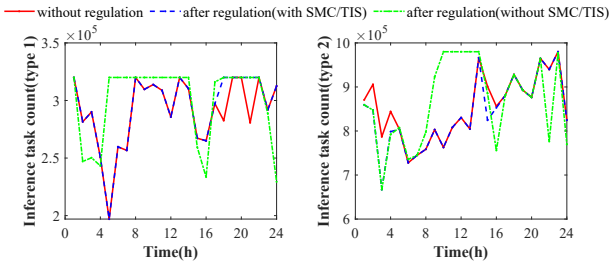



Fig. 6. Impact of economic signals (SMC/TIS) on task volumes at AIDC1 on Day 3.

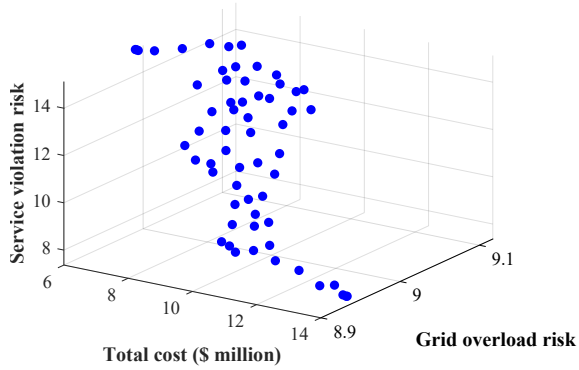



Fig. 7. Pareto fronts of homogeneous task model.

Fig. 6 further shows that, without economic signals, task inflow into AIDC1 frequently approaches the physical service-capacity boundary, thereby converting computational relief into local electrical stress. Therefore, TIS and SMC do not merely modify economic outcomes; they determine whether the identified flexibility is released in an orderly manner.

### *B. Impact of heterogeneous task modeling.*

Fig. 7 shows the Pareto front obtained by the homogeneous task model, while Table IV compares its performance with the proposed heterogeneous model. The homogeneous model achieves a lower $F_1$ of 8.64 million dollars, but the cost reduction comes at the expense of overestimating the feasible workload flexibility. As a result, $F_2$ increases from 8.05 to 9.07, and $F_3$ increases from 8.93 to 11.44. The high $F_3$ region in Fig. 7 further indicates that the homogeneous model cannot effectively protect latency-sensitive workloads under resource constraints. The reason is revealed by the resource allocation results in Table IV(b). Without task classification, $1.09 \times 10^5$ AI task hours are misallocated to CPUs, while $1.19 \times 10^3$ non-AI task hours occupy GPU resources. The GPU/CPU mismatch distorts the dispatchable region and reduces the available capacity for critical inference services. Consequently, as shown in Table IV(c), the $F_3$ of Type-2 inference tasks with the stricter 0.10 s SLA increases to 4.68 under the homogeneous model, whereas the proposed model reduces it to 0.44. These results confirm that heterogeneous task modeling is necessary for separating rigid demand from truly schedulable flexibility and for avoiding overoptimistic flexibility estimation.

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TABLE IV
COMPREHENSIVE COMPARISON OF STRATEGIES: PERFORMANCE, ALLOCATION, AND CRITICAL SHEDDING

| Metric / Item | With Classification (Proposed) | Without Classification (Homogeneous) |
|---|---|---|
| *a. Overall system performance* | | |
| Total cost ($F_1$) [$ M] | 9.12 | 8.64 |
| Overload risk ($F_2$) | 8.05 | 9.07 |
| Service violation risk($F_3$) | 8.93 | 11.44 |
| ESS Cap. (AIDC1/2/3) | 12.00 / 0.79 / 0.31 | 6.58 / 0.20 / 5.38 |
| *b. Resource allocation (GPU/CPU hours)* | | |
| AI tasks on GPU | $2.47 \times 10^6$ | $2.43 \times 10^6$ |
| AI tasks on CPU | 0 | $1.09 \times 10^5$ (Mismatch) |
| Non-AI tasks on GPU | 0 | $1.19 \times 10^3$ (Mismatch) |
| Non-AI tasks on CPU | $8.53 \times 10^5$ | $8.33 \times 10^5$ |
| *c. Service violation risk of AI inference tasks* | | |
| Type 1 (SLA: 0.15s) | 0.80 | 0 |
| Type 2 (SLA: 0.10s) | 0.44 | 4.68 |

### C. Effectiveness of proposed regulation strategies.

To analyze the independent and coordinated effects of ESS and spatiotemporal regulation on the three optimization objectives, four cases are designed with the results summarized in Table V. Taking Case C1 as the baseline reference, the system exhibits the highest grid overload risk, with $F_2 = 25.16$, and a severe service violation risk of $F_3 = 42.37$, confirming that without intervention, AIDC integration causes significant threats to grid security. Case C2 simulates an ESS-only scenario. Although the installation of ESS reduces $F_2$ from 25.16 to 15.74, the service violation risk remains unchanged at 42.37 because the computational bottleneck remains and therefore does not reduce task shedding. Moreover, the strategy requires a large total ESS capacity of 29.33 MWh, leading to the highest $F_1$ of 39.18. In contrast, Case C3 utilizes only internal AIDC flexibility without ESS. This approach reduces $F_2$ and $F_3$ to 9.16 and 9.74 respectively, while maintaining a low $F_1$ of 5.94, indicating that spatiotemporal regulation is the primary mechanism to alleviate overload and task shedding. Case C4 adopts the coordinated strategy. Compared with Case C3, the additional ESS support further reduces $F_2$ from 9.16 to 8.05 and $F_3$ from 9.74 to 8.93, with a moderate increase in $F_1$ to 9.12. Compared with the ESS-only strategy in Case C2, Case C4 achieves better grid security and service quality with a much smaller total ESS capacity of 13.10 MWh. These results indicate that AIDC spatiotemporal regulation serves as the primary mechanism for peak-shaving, while ESS acts as a supplementary tool, effectively enhancing grid security and service quality with only a small cost increase.

TABLE V
PERFORMANCE COMPARISON OF REGULATION STRATEGIES

| Case | Spatiotemporal scheduling | ESS cap. (AIDC1/2/3) | $F_1$ ($M) | $F_2$ | $F_3$ |
|---|---|---|---|---|---|
| C1 | × | × | 7.20 | 25.16 | 42.37 |
| C2 | × | 8.47/19.10/1.76 | 39.18 | 15.74 | 42.37 |
| C3 | ✓ | × | 5.94 | 9.16 | 9.74 |
| C4 | ✓ | 12.00/0.79/0.31 | 9.12 | 8.05 | 8.93 |

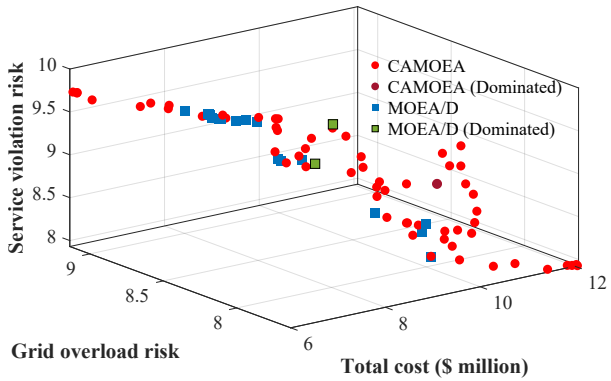



Fig. 8. Convergence and diversity comparison of solution sets: CAMOEA vs. MOEA/D.

### D. Performance comparison of optimization algorithms.

We conduct a comparative study using the proposed heterogeneous task model to evaluate the performance of CAMOEA against the traditional MOEA/D. With the same settings for population size and generations, the Pareto fronts obtained by both algorithms are presented in Fig. 8. Visually, CAMOEA's solution set comprises 60 solutions, while MOEA/D contains only 22. Moreover, the CAMOEA solution set exhibits extensive coverage and uniform distribution in the objective space. MOEA/D's solutions are highly clustered and remain only at the front's edges, struggling to explore the front's interior. The comparative performance initially suggests CAMOEA's superior exploration capability. This indicates the limitation of MOEA/D's fixed weight vectors when handling the irregular and fragmented Pareto front inherent to AIDC scheduling.

Then we use the FMADM method to select the compromise solutions from the Pareto sets, and the results are shown in Table VI. For both algorithms, the results for the three optimization objectives are close. However, the solution diversity of MOEA/D is inferior to that of CAMOEA. To further quantify the differences in solution set quality, the Hypervolume (HV) indicator is employed [39], which evaluates both convergence and diversity. The HV indicator represents the hypervolume formed by the Pareto solution set $X$ and the worst reference point set $P$, with its calculation given by Eq. (52).

$$\mathrm{HV}(X, P) = \bigcup_{x \in X} v(x, P) \tag{52}$$

where U represents the union of hypervolumes formed by each solution x in X and P. According to Eq. (52), CAMOEA's HV indicator is 10.27, superior to MOEA/D's 8.46. Furthermore, we merged the solution sets of both algorithms and statistically analyzed the number and proportion of solutions dominated by the opposing algorithm. The results show that CAMOEA's dominated solution proportion is only 1.67% (1/60), lower than MOEA/D's 9.09% (2/22). This demonstrates that the CAMOEA algorithm generates solutions of higher quality. The dominated solutions for both algorithms are marked in Fig. 8.

### E. Robustness and Feasibility Analysis

*1) Forecast uncertainty robustness:* To examine the influence of imperfect forecasts, the proposed framework is tested

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TABLE VI
CAMOEA VS. MOEA/D REGARDING HV AND DOMINANCE

| Methods | $F_1$ ($M) | $F_2$ | $F_3$ | HV | Dominated ratio |
|---|---|---|---|---|---|
| CAMOEA | 9.12 | 8.05 | 8.93 | 10.27 | $1/60 = 1.67\%$ |
| MOEA/D | 8.17 | 8.42 | 9.14 | 8.46 | $2/22 = 9.09\%$ |

under bounded workload and price prediction errors. Three uncertainty scenarios are considered: $\pm 5\%$ workload forecast errors, $\pm 10\%$ workload forecast errors, and joint $\pm 5\%$ workload and electricity-price forecast errors. The correlated forecast errors are introduced into the deterministic benchmark, and the compromise solution is re-evaluated under each scenario. Table VII summarizes the resulting ESS capacities and objective values. Compared with the deterministic benchmark, forecast errors lead to moderate increases in $F_1$ and $F_3$, while $F_2$ remains relatively stable. For example, under the $\pm 10\%$ workload-error scenario, $F_1$ increases from 9.12 to 9.35 and $F_3$ increases from 8.93 to 9.54, whereas $F_2$ remains within a comparable range. These results indicate that the deterministic benchmark remains a useful reference for planning, although uncertainty introduces measurable performance degradation.

TABLE VII
NUMERICAL OUTCOMES UNDER FORECAST ERROR SCENARIOS

| Scenario | ESS cap. (MWh) | $F_1$ | $F_2$ | $F_3$ |
|---|---|---|---|---|
| Deterministic | 12.00/0.79/0.31 | 9.12 | 8.05 | 8.93 |
| $\pm 5\%$ workload | 10.18/0.67/2.64 | 9.15 | 7.99 | 8.96 |
| $\pm 10\%$ workload | 6.71/4.31/3.35 | 9.35 | 7.81 | 9.54 |
| $\pm 5\%$ workload/price | 11.79/2.55/0.25 | 9.39 | 7.96 | 9.22 |

*2) PV-integration extension:* An on-site PV profile is incorporated into the proposed framework. The total ESS capacity decreases from 13.10 MWh to 7.19 MWh, while the three objectives improve from $9.12/8.05/8.93$ to $7.20/7.23/7.81$, confirming that the proposed coordination mechanism remains effective when local renewable generation is considered.

*3) SOCP-relaxation tightness:* Since the distribution network constraints are represented by the SOCP relaxation of the DistFlow equations, the tightness of the relaxation should be verified to ensure the physical feasibility of the obtained scheduling solutions. For each branch $ij$ and time slot $t$, the SOCP relaxation gap is defined as Eq. (53), then the maximum relaxation gap is calculated as (54).

$$\varepsilon_{ij,t} = l_{ij,t} - (P^2_{ij,t} + Q^2_{ij,t})/v_{i,t}, \quad \forall ij \in \mathcal{L}, t \in \mathcal{T} \tag{53}$$

$$\varepsilon^{max} = \max_{ij \in \mathcal{L}, t \in \mathcal{T}} |\varepsilon_{ij,t}| \tag{54}$$

After solving the proposed coordinated scheduling problem, the relaxation gaps of all Pareto optimal solutions are checked through (53) and (54). As shown in Fig. 9, the maximum observed gap is $5.63 \times 10^{-5}$ p.u., which is numerically negligible. This result indicates that the SOCP relaxation remains tight under the obtained AIDC scheduling and ESS operation solutions. Therefore, the reported power-flow results are physically feasible.

*4) Scalability analysis:* The computational performance of the proposed planning framework is evaluated in terms of wall-clock runtime and parallel scalability. Across 8 independent CAMOEA runs with identical parameter settings, the average runtime was 37.6 h, with a standard deviation of 1.6 h. The computational cost mainly arises from the repeated solution of the embedded operational optimization problems. The fitness evaluations of different individuals within the same generation are mutually independent and can therefore be executed in parallel. A modern multicore processor, computational workstation, or high-performance computing platform can solve multiple embedded CPLEX problems concurrently. Together with parallel fitness evaluation, such computing resources can substantially reduce the computational time. For example, assuming near-linear parallel scaling and limited communication overhead, increasing the number of parallel evaluations from 4 to 24 would reduce the estimated runtime from 37.6 h to approximately $37.6 \times 4/24 \approx 6.3$ h.

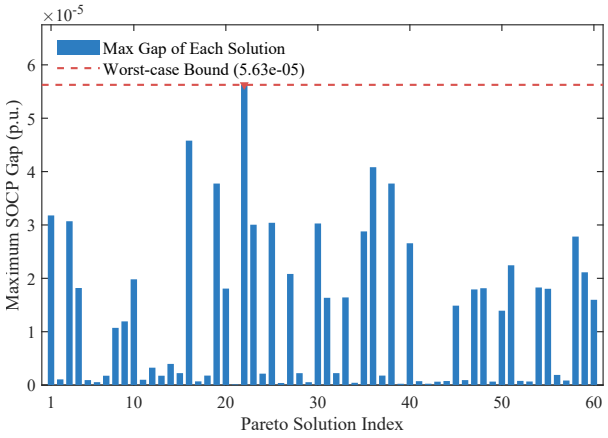



Fig. 9. Maximum SOCP relaxation gaps across all Pareto optimal solutions.

## F. Sensitivity Analysis

A sensitivity analysis is conducted for the two key parameters that directly control spatiotemporal scheduling behavior: the TIS pricing coefficient $\alpha_k$ and the SMC weighting coefficient $\beta_k$.

*1) One-dimensional sensitivity analysis:* Set the baseline values of $\alpha_k$ and $\beta_k$ as 1.0, and vary them independently to observe their impact on the three objectives ($F_1$, $F_2$, $F_3$). The change ratio of $\alpha_k$ is varied from 0.5 to 4.0, while $\beta_k$ is varied from 0.5 to 1.5.

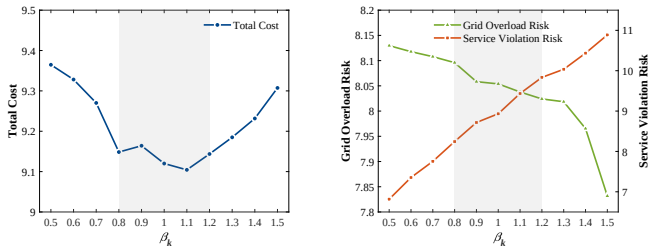
Fig. 10 illustrates the sensitivity of the objectives to changes in $\beta_k$. When $\beta_k$ is relatively small, the penalty for spatial scheduling is insufficient, leading to excessive and uncoordinated inter-AIDC workload migration. As a result, communication overhead increases, and part of the migrated load may be redirected to electrically stressed buses, thereby raising the grid overload risk $F_2$. In contrast, an overly large $\beta_k$ strictly constrains spatial flexibility of the system. To mitigate local grid congestion, the system is compelled to implement task shedding, increasing the service violation risk $F_3$. Moreover, a high $\beta_k$ prevents workloads from utilizing surplus grid capacity at other AIDCs during off-peak periods. The resulting local capacity saturation weakens temporal workload shifting and reduces $R^{TIS}$, causing a secondary rebound in the total cost when $\beta_k \geq 1.1$.

With $\beta_k$ fixed at 1.0, Table VIII reports the sensitivity results with respect to $\alpha_k$. As $\alpha_k$ increases from 0.5 to 4.0, the

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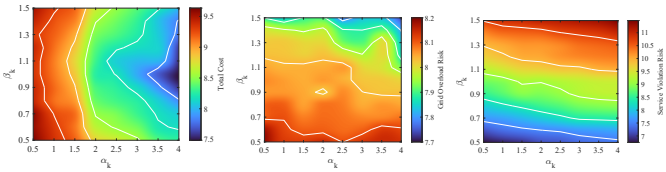



(a) (b)

Fig. 10. Sensitivity analysis of $\beta_k$: (a) total cost; (b) grid overload risk and service violation risk.

TABLE VIII
SENSITIVITY OF OBJECTIVES TO $\alpha_k$

| $\alpha_k$ | Total cost $F_1$ | Grid overload risk $F_2$ | Service violation risk $F_3$ |
|---|---|---|---|
| 0.5 | 9.359 | 8.068 | 8.847 |
| 1.0 | 9.120 | 8.053 | 8.933 |
| 1.5 | 8.858 | 8.043 | 9.161 |
| 2.0 | 8.273 | 8.056 | 9.130 |
| 2.5 | 8.198 | 8.041 | 9.228 |
| 3.0 | 7.972 | 8.022 | 9.315 |
| 3.5 | 7.715 | 8.018 | 9.383 |
| 4.0 | 7.490 | 8.004 | 9.386 |



(a) (b) (c)

Fig. 11. Two-dimensional sensitivity analysis of $\alpha_k$ and $\beta_k$: (a) total cost $F_1$; (b) grid overload risk $F_2$; (c) service violation risk $F_3$.

total cost $F_1$ decreases monotonically, reflecting the stronger incentive to exploit TOU price differences and increase $R^{TIS}$. Meanwhile, the grid overload risk $F_2$ remains stable, with a maximum deviation of less than 0.61% from the baseline case $(\alpha_k, \beta_k) = (1, 1)$. Such stability suggests that $\alpha_k$-guided temporal shifting mainly utilizes available off-peak capacity without aggravating local overload. However, the service violation risk $F_3$ gradually increases as $\alpha_k$ becomes larger. The underlying reason is that excessive temporal concentration of workloads may saturate the hosting capacity of local AIDC buses, which further limits inter-AIDC migration flexibility and increases the risk of service degradation.

*2) Two-dimensional sensitivity analysis:* Fig. 11 further presents a 2D sensitivity analysis of the coupling effects between $\alpha_k$ and $\beta_k$. For $F_1$, a larger $\alpha_k$ generally reduces the total cost by increasing $R^{TIS}$, while an excessively small $\beta_k$ causes frequent migrations and higher communication costs. An overly large $\beta_k$ suppresses spatial flexibility, leading to local task accumulation and weaker temporal arbitrage. Therefore, the lowest-cost region appears around $\beta_k \approx 0.8$–$1.2$. The grid overload risk $F_2$ changes only slightly across the tested parameter space, indicating that the proposed model maintains grid security under different economic-parameter settings. A mild increase of $F_2$ can be observed in the low-$\beta_k$ region, where insufficient migration penalties may redirect tasks toward capacity-rich but electrically stressed nodes. For $F_3$, service violation risk increases when spatial flexibility is overly restricted or when a high $\alpha_k$ concentrates flexible workloads in low-price periods. Overall, moderate values of $\alpha_k$ and $\beta_k$ provide a balanced trade-off among $F_1$, $F_2$, $F_3$.

Based on the tested cases, the recommended ranges are selected as $1.0 \leq \alpha_k \leq 2.0$ and $0.8 \leq \beta_k \leq 1.2$. Within this range, the model maintains stable grid overload performance while avoiding excessive service degradation or unnecessary spatial migration.

## VI. CONCLUSION

This paper proposes a coordinated framework for AIDC-Grid scheduling and ESS planning to address the critical challenges imposed by the integration of AIDC with heterogeneous workloads. By introducing a refined heterogeneous load model and dual economic indicators (TIS/SMC), the proposed method achieves a trade-off among operational cost, grid security, and service quality. Simulation results on a modified IEEE 33-bus system demonstrate that the homogeneous approach fails to capture the rigidity of latency-sensitive inference tasks, resulting in a severe service violation risk of 11.44. In contrast, the proposed heterogeneous modeling is indispensable for correcting the "flexibility overestimation" inherent in homogeneous approaches, and significantly reduces the service violation risk to 8.93, while reducing service violations under strict SLAs. Furthermore, the TIS and SMC indicators transform AIDCs from passive consumers into active flexible grid assets, where spatiotemporal regulation acts as the primary measure for congestion relief, while ESS serves as a complementary buffer. The proposed method reduces the annualized total system cost to \$9.12 million, representing a 76.72% reduction compared to the \$39.18 million required for an ESS-only approach. Robustness tests under bounded forecast errors further confirm that the main conclusions remain valid when realistic prediction errors are incorporated. Additionally, the algorithmic performance analysis validates the necessity of the introduced CAMOEA, which outperforms MOEA/D in terms of Hypervolume (10.27 vs. 8.46) when handling irregular Pareto fronts derived from discrete scheduling. Future work will further consider stochastic renewable generation and explore multi-energy coupling mechanisms to further enhance the resilience and sustainability of AIDCs.

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